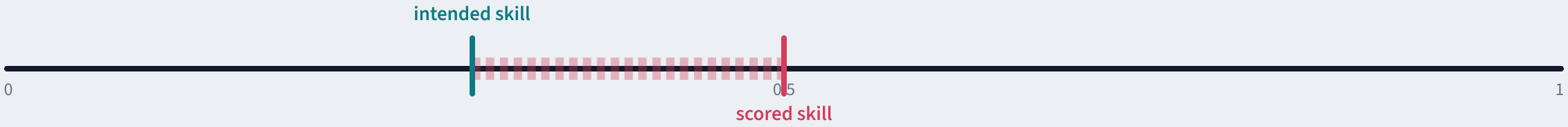
Can a ruler check its own markings?

Digital twins of students, and how to calibrate assessments and assessment bots with them.



30-minute workshop · talk, demo, then hands-on with a web chatbot

Generated items can't be calibrated the old way.

Classical calibration

Fixed item bank. Each item gets difficulty and discrimination estimates from **50–200 real responses** before use.

LLM-generated assessment

Every student can get a different item. The rubric changes each semester. There is no time to collect 50 responses per item.

The bootstrapping problem: you can't validate without student data, and you can't deploy without validation. Simulated students fill the gap.

Prior work reports LLM-simulated respondents approximating human item calibration with correlations above 0.89 (Liu 2025). The catch is on the next slide.

Four things simulated students can do. We focus on two.

Teacher training

Practise diagnosing and responding to common misconceptions on a student who never gets tired.

Item design

Check whether a question and its rubric separate weak from strong answers, before a real student sees it.

Today

Whole classroom

Simulate a mixed class to plan pacing, grouping and how a lesson lands across levels.

Tutor development

Stress-test a tutor or assessment bot: does it ask the right next question, and route each student fairly?

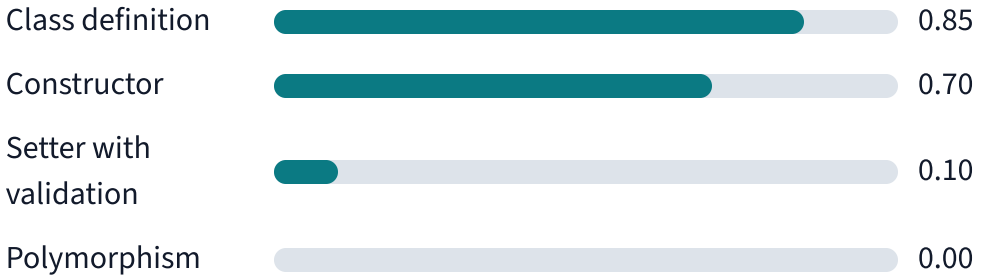
Today

Both of today's uses share one question: **when the twin gets a score, is the score right?**

A digital twin of a student is a skill profile an LLM can act out.

The profile

A vector of skills, each from 0 (absent) to 1 (mastered). Here, 4 of the 24 Python OOP skills in the paper.



The behaviour

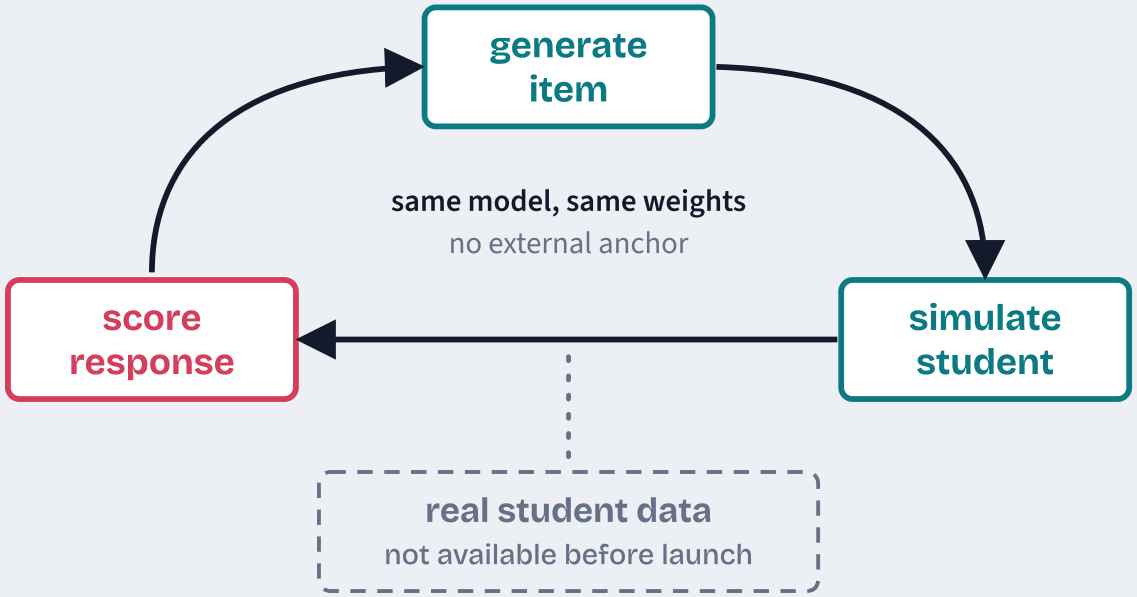
Prompt the model to answer a task as that student, mistakes included.

```
class Account:
    def __init__(self, owner, bal):
        self.owner = owner
        self.balance = bal   # no setter, no check

    def deposit(self, amt):
        self.balance = self.balance + amt
```

Simulated students are cheap, repeatable and available before any real student has seen the item.

One LLM writes the item, plays the student, and grades the answer.



It plays weak students badly

Expert knowledge leaks through, and none of 11 LLMs matched real student score distributions.

Yuan 2026; Srivatsa 2025

It favors its own writing

LLMs rate their own outputs above equivalent text from other sources.

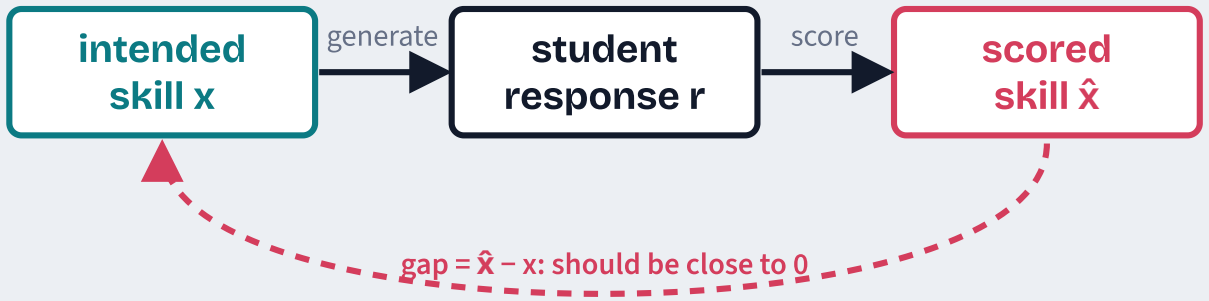
Panickssery 2024

It disagrees with itself

Even proprietary models score inconsistently at temperature above 0.

Wang 2024

Adaptive drift is the gap between what we asked for and what we scored.



E[score(generate(*x*))]

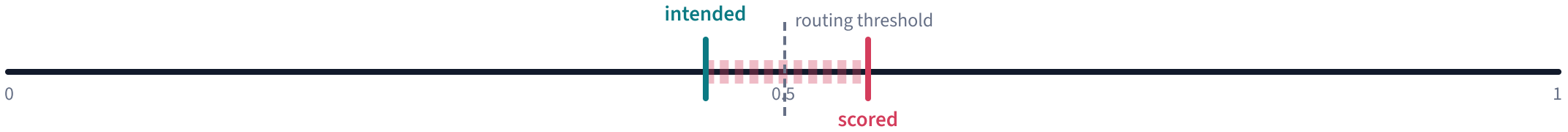
≈ *x*

Generative-Evaluative Agreement (GEA)

Small drift near the threshold sends a student down the wrong path.

Drag the student's intended skill

0.45



Drift (scored − intended)	Should be routed to	Actually routed to
+0.10	Low path	High path

Schematic shaped after Figure 4 of the paper (weak skill overscored by about +0.20, converging near the diagonal above 0.8). Not a fit to the raw data.

Four levers narrow the gap. You'll pull some of them in a minute.

Break the rubric into skills

Score each skill separately, with a clear line between "has it" and "doesn't", then average. The paper's main proposal.

Use a different model to score

A model rates its own writing higher. Let another one grade.

Say which misconceptions are active

"Act like a beginner" leaks expertise. "Never validates input, believes class variables are per-object" doesn't.

Check against a few real students

10–20 real answers anchor the simulated numbers.

Next: a live demo of the drift, then you try it with a chatbot in your browser.

Source: Lee, Wang, Lye & Peh, "Generative-Evaluative Agreement: A Necessary Validity Criterion for LLM-Enabled Adaptive Assessment", BEA 2026.